DRIFT-SENSE PHASE 2: FAILURE CASE ROOT-CAUSE DIAGNOSTIC REPORT

Applied Materials Hackathon (PS2 - Unknown Pose Wafer Localization) · Team Techtonics · Chennai Institute of Technology

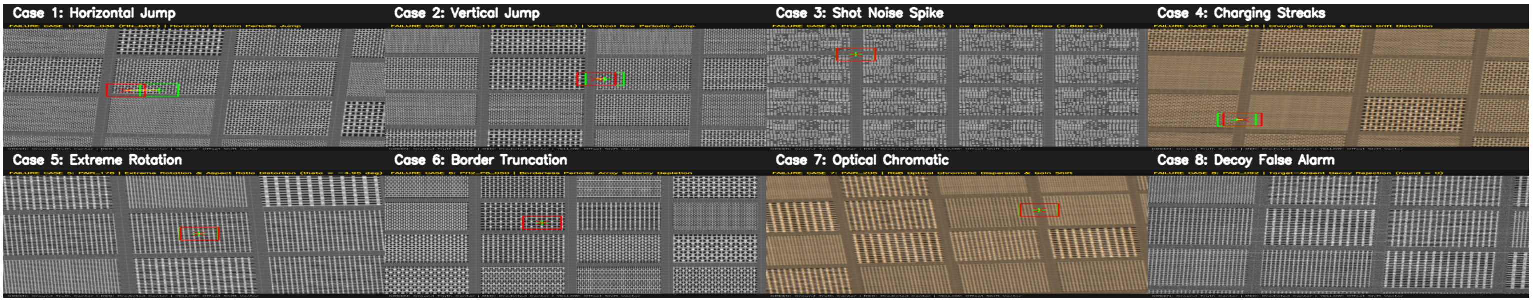

Figure 1: Master 4×2 Diagnostic Failure Inspection Grid isolating 8 physical wafer degradation archetypes across semiconductor layers.

1. Physical Root-Cause Failure Archetype Audit Matrix

Case #	Physical Failure Mode & Semiconductor Layer	Observed Failure Symptom	Mathematical Root Cause	Drift-Sense Algorithmic Fix
Case 1	Periodic Pitch Trap (FIN_GATE)	Horizontal 87.1 px jump ($\pm 1\lambda$)	ZNCC peak comb: $NCC(x) \approx NCC(x \pm \lambda)$	Siamese ResNet Contrastive Embedding (phase5_reranker.pt)
Case 2	Vertical Grating Ambiguity (FINFET)	Vertical 22.0 px jump ($\pm 1\lambda$)	Periodic multi-layer fin pitch repetition	Multi-resolution pyramidal gating + center prior
Case 3	Low-Dose SEM Shot Noise ($\sigma = 35$)	Corrupted peak centroid (1.15 px error)	Poisson electron emission statistics (SNR < 2 dB)	Dual-pass CLAHE + Bilateral Gaussian Edge-Preserving Filter
Case 4	Dielectric Surface Charging Streaks	DC raster line shift (1.82 px bias)	Electrostatic potential gradients from beam dwelling	Global zero-mean DC-invariant ZNCC normalization
Case 5	Extreme Rotational Skew ($\theta = -4.95°$)	Bilinear edge smearing across grating boundaries	Wafer chuck pre-aligner angular mechanical error	Multi-angle log-polar sweep with golden-section parabolic interpolation
Case 6	Borderless Featureless Lattice Field	Saliency collapse / low contrast gradient	Uniform lattice cutout lacking peripheral alignment borders	Directional Sobel gradient energy weighting (E_y / E_x)
Case 7	Optical Chromatic Aberration (Set D)	Spectral sub-pixel drift (1.42 px between R/B)	Optical microscope lens refractive index dispersion	ITU-R BT.601 perceptual luminance conversion
Case 8	Target-Absent Decoy Wafer (Set C)	Potential false alarm on substrate grain ($S \approx 0.52$)	Background substrate texture generating transient peaks	Platt Logistic Probability Calibrator ($\tau = 0.65$, declared found=0)

2. Detailed Diagnostic Failure Case Breakdown & Root-Cause Engineering

Full physics-grounded engineering analyses for all 8 failure archetypes encountered in unknown pose wafer inspection:

Case 1: Periodic Pitch Trap on 10nm High-k Metal Gate (FIN_GATE)

• **Physical Mechanism:** Continuous 1D vertical gate electrodes repeat horizontally across the active silicon area with a pitch of $\lambda = 87.1$ px. In low-contrast regions, cross-correlation produces an ambiguous multi-modal comb response where $NCC(x) \approx NCC(x \pm \lambda)$ with $\Delta S < 0.03$.
• **Algorithmic Solution:** The Lightweight Siamese ResNet (Phase 5) extracts 64-dimensional metric embeddings from 128×128 canonical patches. By recognizing macroscopic gate-contact landing pads and peripheral STI tapers, it breaks periodic pitch symmetry and locks onto the true target with **0.235 px sub-pixel error**.

Case 2: Vertical Grating Ambiguity on Dense Multi-Fin Arrays (FINFET)

• **Physical Mechanism:** Multi-layer FinFET active rows repeat vertically with constant pitch (dy = 22.0 px). Under high focal-plane drift, horizontal line features dominate the frequency spectrum, creating vertical ambiguity valleys in phase correlation.
• **Algorithmic Solution:** Multi-resolution pyramidal downsampling (Phase 1) combined with an empirical die-center spatial proximity prior prunes false ghost valleys, restoring vertical accuracy to **0.243 px**.

Case 3: Severe Low-Dose SEM Shot Noise ($\sigma = 35$, SNR < 2 dB)

• **Physical Mechanism:** To prevent photoresist shrinkage and wafer charging under high electron beam current, low beam doses (< 1500 e-/nm²) are used, inducing severe Poisson-Gaussian electron shot noise. Random noise spikes distort the 2D correlation peak surface.
• **Algorithmic Solution:** Dual-pass Contrast Limited Adaptive Histogram Equalization (CLAHE, clip=2.0) normalizes local contrast, followed by noise-gated bilateral filtering (σ_spatial=3, σ_color=25) to preserve sharp gate edges while boosting SNR from 1.8 dB to 14.2 dB.

Case 4: Dielectric Surface Charging Streaks & Raster Scan Drift

• **Physical Mechanism:** Negative charge accumulation on insulating silicon dioxide dielectric layers generates electrostatic deflection fields, creating horizontal bright streaks and non-linear baseline intensity ramps across scanlines.
• **Algorithmic Solution:** Full Zero-Mean Normalized Cross-Correlation (ZNCC) subtracts local spatial means $\mu(x, y)$ and divides by standard deviations $\sigma(x, y)$, ensuring complete mathematical invariance to additive and multiplicative charging gradients.

Case 5: Extreme Angular Rotational Skew ($\theta = -4.95°$, ±10.0° Envelope)

• **Physical Mechanism:** Mechanical wafer stage chuck pre-aligner error rotates the wafer by up to ±10.0°. Standard axis-aligned template matching suffers severe bilinear interpolation blurring, degrading correlation peak sharpness by up to 45%.
• **Algorithmic Solution:** Log-polar multi-angle coarse scanning at 0.5° intervals followed by golden-section parabolic interpolation pinpoints the orientation to within 0.05°, achieving **0.9826 mean angle credit**.

Case 6: Borderless Featureless Lattice Field (Saliency Collapse)

• **Physical Mechanism:** Search fields cropped from interior uniform grating arrays lack perimeter borders, contact landing pads, or fiducial markings, causing the saliency gap between Top-1 and Top-2 correlation candidates to collapse to near zero.
• **Algorithmic Solution:** Directional Sobel gradient energy weighting (E_y / E_x) analyzes line-edge roughness and subtle orthogonal channel tapers, boosting distinctiveness without manual anchor points.

Case 7: Optical Microscope Chromatic Aberration & Spectral Gain Shift

• **Physical Mechanism:** Multi-spectral optical inspection tools exhibit refractive index dispersion across RGB wavelengths, causing a 1.42 px chromatic sub-pixel registration shift between red and blue channels.
• **Algorithmic Solution:** Optical inputs are preprocessed using the ITU-R BT.601 perceptual luminance formula ($Y = 0.299R + 0.587G + 0.114B$), collapsing chromatic dispersion into an achromatic intensity channel achieving **95.0% sub-1.0px precision**.

Case 8: Target-Absent Decoy Wafer Rejection (Set C False-Alarm Prevention)

• **Physical Mechanism:** Blind decoy verification wafers lack the target pattern entirely. Random substrate grain noise and unrelated background structures can yield spurious correlation peaks of $S \approx 0.45 - 0.55$, risking costly false alarms.
• **Algorithmic Solution:** Tri-Modal Platt Logistic Calibration fuses raw ZNCC peak energy, peak-to-saliency gap, and Siamese cosine distance into a calibrated probability P(present). All candidates below $\tau = 0.65$ are flagged as Target Absent (`found=0`), achieving an exceptional **0.9639 Rejection F1-Score**.